Written Critique Assignment 2

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1 QUESTIONS ABOUT WHAT-IF TOOL

1.1 Question 1

According to the Google for Developers Machine Learning Glossary (2026), a false negative is "an example in which the model mistakenly predicts the negative class," a false positive is when a model makes an "incorrect prediction of the positive class." While accuracy is "the number of correct classification predictions divided by the total number of predictions" (Google for Developers, 2026).

1.2 Question 2

Bias is known for concepts like "stereotyping, prejudice or favoritism towards some things, people, or groups over others" (Google for Developers, 2026). In this dataset, bias shows up because the tool tends to falsely flag black defendants as future criminals, almost twice at the rate as white defendants (Angwin et al., 2016). Fairness is the active effort to fix this issue for fair treatment. The main difference is that bias is the unfair mistake built into the system, while fairness is the fix to make things right.

1.3 Question 3

To reduce bias in this tool, false positive rate should be the main focus. The biggest problem with this system is that it is biased against one group, as "black defendants were still 77 percent more likely to be pegged as at higher risk of committing a future violent crime" even when they never committed another crime (Angwin et al., 2016). Because of this, innocent black people to face consequences like longer jail times, lowering this false positive rate is the most direct way to reduce the bias.

1.4 Question 4

To ensure fairness, false positive is still the most important factor. True fairness means that the chance of being wrongly accused should be the same for everyone, ensuring everyone gets the fair treatment across different races. Since the worst thing this tool does is wrongly label black people as dangerous, fairness means making sure no single racial group has to deal with that mistake more than other group.

1.5 Question 5

Yes, they are. Because the main ethical problem with this tool is that it gives out way too many false positives to one specific group. Researchers note that when a system makes biased mistakes that harm one demographic group over another, fixing the underlying bias and making the system fair both require balancing out those specific error rates by normalizing the false positive rate (Chouldechova, 2017).

1.6 Question 6

Feature Value	Count	Threshold	False Positives (%)	False Negatives (%)	Accuracy (%)	F1
African-American	5338	0.85	8.0	33.3	58.8	0.49
Caucasian	3359	0.62	8.2	26.8	65.0	0.46
Hispanic	739	0.63	6.2	26.0	67.8	0.35
Other	493	0.4	6.5	25.8	67.7	0.43
Asian	37	0.38	13.5	21.6	64.9	0.24
Native American	34	0.92	0.0	32.4	67.6	0.52

Figure 1—The result after changing the threshold

By manually increasing the threshold for African-American defendants from 0.66 to 0.85 in Figure 2, the false positive rate came down from 19.1% to an 8.0% match with the Caucasian baseline. However, there is trade-offs required to achieve this, such as the false negative rate for African-Americans drastically jumps from 15.9% to 33.3%, and their overall accuracy drops from 65.0% to 58.8%. This shift negatively impacts the broader community because the model will now incorrectly classify twice as many actual re-offenders as low-risk, proving that fixing the false positive bias directly degrades the model's ability to catch true risks.

1.7 Question 7

As demonstrated in the previous analysis, mathematically forcing fairness by equalizing the false positive rates across all racial groups negatively affected the other performance metrics. Specifically, balancing the false positives forces the false negative rate for African-American defendants to spike to 33.3% and causes their overall accuracy to drop to 58.8%. These mathematical trade-offs negatively impact the broader community causing more unfairness.

1.8 Question 8

Yes, it is quite difficult to mitigate bias and ensure fairness simultaneously across all performance metrics. As demonstrated above when manually adjusting the thresholds in the What-If Tool, forcing the false positive rates to be equal successfully corrected one specific bias but completely impacted the rest of the model. Lowering the false positive rate for African-American defendants caused their false negative rate to drastically spike to 33.3% and dropped their overall accuracy to 58.8%. Researchers have proven mathematically that it is impossible for a system to achieve equal accuracy, equal false positives, and equal false negatives all at the same time unless the historical base rates are perfectly identical across groups (Chouldechova, 2017). Because fixing one unfair mistake inherently breaks another part of the system, creating a fair model will requires humans to make ethical trade-offs about which types of errors are the least harmful.

1.9 Question 9

These same ideas apply to any other dataset because the problem of balancing false positives, false negatives, and accuracy is always the same. However, the specific number you care about most will change depending on what the tool is doing, because what is fair depends on context and the specific application. For example, if a tool is looking for cancer, a false negative of missing the cancer entirely is much more dangerous than a false positive a scary false alarm, so you would completely change which mistake you want to fix first.

REFERENCES

[1] Angwin, J., Larson, J., Mattu, S., & Kirchner, L. (2016, May 23). "Machine bias." *ProPublica*. URL: https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing.

[2] Chouldechova, A. (2017). "Fair prediction with disparate impact: A study of bias in recidivism prediction instruments." *Big Data*, 5(2), 153-163.

[3] Google for Developers. (2026, April 13). "Machine learning glossary." *Google*. URL: https://developers.google.com/machine-learning/glossary.